

Applications of Monotone Operator Theory

Chapter 8: Integral Equations and Nonlinear Analysis

From: An Introduction to Nonlinear Analysis and Fixed Point Theory (Pathak, 2018)

Pages 641–664

Chapter Overview

- **Focus:** Applications of monotone operator theory to integral equations
- **Main Topics:**
 - 1 Integral Equations (Hammerstein, Urysohn, generalized types)
 - 2 Operator theoretic approach to integral equations
 - 3 Existence and uniqueness theorems
 - 4 Applications to nonlinear problems
 - 5 Computational schemes
- **Framework:** Hilbert and Banach spaces with monotone operators
- **Goal:** Develop solvability results using fixed point theory

Types of Integral Equations

1 Nonlinear Hammerstein integral equation:

$$x(s) + \int_{\Omega} k(s, t) f(t, x(t)) dt = y(s)$$

2 Nonlinear integral equation of mixed type:

$$x(s) + \int_{\Omega} k_1(s, t) f_1(t, x(t)) dt + \int_{\Omega} k_2(s, t) f_2(t, x(t)) dt = y(s)$$

3 Generalized Hammerstein integral equation:

$$x(s) + \int_{\Omega} k_v(s, t) f(t, x(t)) dt = y(s)$$

where k_v is a kernel depending on the solution.

4 Urysohn's integral equation:

$$x(s) + \int_{\Omega} k(s, t, x(t)) dt = y(s)$$

Hammerstein Integral Equation

Hammerstein Integral Equation

$\Omega \subseteq \mathbb{R}^n$ measurable

$K : \Omega \times \Omega \rightarrow \mathbb{R}$ kernel function

$f : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ Carathéodory

$$x(s) + \int_{\Omega} k(s, t) f(t, x(t)) dt = y(s)$$

$[Kx](s) = \int_{\Omega} k(s, t) x(t) dt$ (linear operator)

$[N_f x](s) = f(s, x(s))$ (Nemytskii operator)

Operator Equation Reduction

Integral Equation

$$x(s) + \int_{\Omega} k(s, t) f(t, x(t)) dt = y(s)$$



*Reduction to
Operator Form*

$$x + KN_f x = y$$

Operator Equations

$$\text{Linear: } [Kx](s) = \int_{\Omega} k(s, t) x(t) dt$$

$$\text{Nonlinear: } [N_f x](s) = f(s, x(s))$$

$$\text{Composed: } x + KN_f x = y$$

Solve using operator theoretic methods:

monotone operators, fixed points, Banach spaces

Theorem 8.8: Noncompact Case

Theorem (Dolph-Minty)

Let $\mathcal{H}$ be a Hilbert space and $N_f : \mathcal{H} \rightarrow \mathcal{H}$ be hemicontinuous and monotone. Let $K : \mathcal{H} \rightarrow \mathcal{H}$ be continuous and strongly monotone.

Then equation (8.31) has a unique solution x for every y , and this solution x depends continuously on y and satisfies:

$$\|x\| \leq \frac{1}{c} \|K\| \|y\|$$

where c is the constant of strong monotonicity.

Key Conditions:

- K strongly monotone: $(Kx, x) \geq c\|x\|^2$ for some $c > 0$
- N_f hemicontinuous and monotone
- Solution depends continuously on data

Example 8.6: Hammerstein Integral Equation

Problem:

$$x(s) + \int_{\Omega} k(s, t) f(t, x(t)) dt = 0$$

where Ω is arbitrary, $k : \Omega \times \Omega \rightarrow \mathbb{R}$ is continuous symmetric positive semidefinite.

Assumptions on f :

- ① $f(\cdot, u)$ is measurable on Ω for all $u \in \mathbb{R}$
- ② $f(s, \cdot)$ is continuous on $\mathbb{R}$ for almost all $s \in \Omega$
- ③ There exist $\rho > 0$ and constants a, b, M such that:
 - $|f(s, x)| \leq a(s) + b|x|$, $a \in L_2(\Omega)$
 - $f(s, x) \geq a + bx$ for $x > M$
 - $f(s, x) \leq -a + bx$ for $x < -M$

Example 8.6 (continued): Solution

Operator Equation Reformulation:

$$x + KN_fx = 0$$

Analysis:

- $K : L_2(\Omega) \rightarrow L_2(\Omega)$ is bounded linear monotone operator
- $N_f : L_2(\Omega) \rightarrow L_2(\Omega)$ is continuous bounded monotone operator
- By Theorem 8.9, the operator equation has a solution
- The growth conditions on f ensure coercivity: $(N_fx, x) \geq -\varphi_0$ for some φ_0

Conclusion:

The Hammerstein integral equation has a solution in $L_2(\Omega)$

Angle Bounded Linear Operators

Angle Bounded Linear Operators

Definition: $K : X \rightarrow X^*$ is angle bounded with constant $c \geq 0$ if:

$$|(Kx_1, x_2) - (Kx_2, x_1)| \leq 2c \sqrt{(Kx_1, x_1)(Kx_2, x_2)}$$

Properties:

- Monotone: $(Kx, x) \geq 0$ for all $x \in X$
- Symmetric: $K^* = K$

Examples: sector: numerical range in a sector

Definition 8.2: Angle Bounded Operator

Definition

Let X be a Banach space with dual X^* . A bounded linear operator $K : X \rightarrow X^*$ is said to be **angle bounded** with constant $c \geq 0$ if for all $x_1, x_2 \in X$:

$$|(Kx_1, x_2) - (Kx_2, x_1)| \leq 2c \sqrt{(Kx_1, x_1)(Kx_2, x_2)}$$

Geometric Interpretation:

- K is **monotone**: $(Kx, x) \geq 0$ for all x
- Numerical range lies in a sector of angle determined by c
- K is **symmetric**: $K^* = K$ (hence angle bounded with $c = 0$)
- Special case: $(Kx, x) = (Kx, x)$ (self-adjoint duality)

Properties of Angle Bounded Operators

Important Properties:

- 1 If K is angle bounded, then K is monotone
- 2 The spectrum of K lies in a sector determined by the angle bound c
- 3 If K is completely continuous and angle bounded, then fixed point theorems apply
- 4 Angle boundedness is preserved under perturbations by compact operators

Applications:

- Variational methods for nonlinear PDEs
- Fredholm theory with perturbations
- Bifurcation theory
- Spectral theory in Banach spaces

Link to Integral Equations:

If $K(s, t)$ is symmetric, then the integral operator Kx is monotone and angle bounded (with $c = 0$).

Generalized Hammerstein Equation

A nonlinear integral equation of generalized Hammerstein type is:

$$u(s) + \int_{\Omega} k(s, t, u(t))f(t, u(t))dt = v(s)$$

where the kernel $k(s, t, u)$ **depends on the solution** u .

Operator Reformulation:

For a fixed $u \in X$, the integral operator $K(u)$ is defined as:

$$[K(u)x](s) = \int_{\Omega} k(s, t, u(t))x(t)dt$$

The generalized equation becomes:

$$u + K(u)N_f u = v$$

This requires $K(u)$ to be well-defined and satisfy appropriate monotonicity conditions for each u .

Theorem 8.17: Generalized Hammerstein

Theorem

Let X be a real reflexive Banach space, and for each $u \in X$, let $K(u) : X^ \rightarrow X$ be a linear angle-bounded operator with constant $c \geq 0$.*

Assume $K(u)$ is completely continuous and $u \mapsto K(u)$ is continuous from X to $L(X^, X)$.*

Further, assume $\|K(u)\| \leq r$ for each $u \in X$. Let $N_f : X \rightarrow X^$ be continuous and bounded satisfying:*

$$(u_1 - u_2, N_f u_1 - N_f u_2) \geq -k \|u_1 - u_2\|^2$$

for $kr < 1 + c^2$.

Then the equation $u + K(u)N_f u = 0$ has at least one solution in X .

Key Innovation: Depends on Backwinkel-Schillings technique and fixed point method.

Urysohn's Integral Equation

Urysohn's Integral Equation

$$x(s) + \int_{\Omega} k(s, t, x(t)) dt = y(s)$$

Generalization: kernel depends on solution

$k = k(s, t, x)$ (not just $k(s, t)$)

Key Features:

- Nonlinear integral equation
- Kernel depends on solution value
- Reduces to Hammerstein for $k = k(s, t) \cdot f(t, x(t))$
- Applications: physics, mechanics, biology

Theorem on Urysohn's Equation

Urysohn's Equation:

$$x(s) + \int_{\Omega} k(s, t, x(t)) dt = y(s)$$

Existence Result:

Under appropriate conditions on k :

- If $k(s, t, x)$ is Carathéodory and satisfies growth bounds
- If $\int_{\Omega} |k(s, t, x_0)| ds$ is bounded
- If monotonicity conditions hold

Then:

$$x(s) + \int_{\Omega} k(s, t, x(t)) dt = y(s) \quad \text{has a solution in } L_2(\Omega)$$

Proof Strategy:

- 1 Reformulate as operator equation $x + Nx = y$
- 2 Show N is continuous, bounded, and monotone
- 3 Apply fixed point theorem or Leray-Schauder principle
- 4 Use Banach contraction or monotone operator theory

Computational Methods

Projection Methods:

For approximate solutions, use finite-dimensional approximations:

$$\text{Let } X_n \subseteq X_n \subseteq \dots \rightarrow X$$

$$P_n : X \rightarrow X_n \quad (\text{projection operators})$$

Approximate Equation:

$$u_n + P_n K(u_n) N_f P_n u_n = P_n v$$

Convergence:

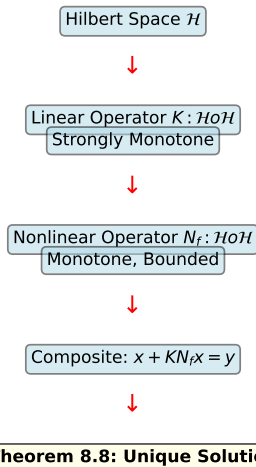
- Strong convergence: $u_n \rightarrow u$ in norm
- Weak convergence: $u_n \rightharpoonup u$ weak topology
- Error estimates: $\|u - u_n\|$ bounds

Iterative Methods:

- Mann iteration
- Ishikawa iteration
- Gradient descent methods
- Galerkin approximations

Monotone Operators Framework

Monotone Operators Framework



Summary of Main Results

Central Theme: Integral equations reduce to fixed point problems

Equation Type	Operator Form	Solvability
Hammerstein	$x + KN_f x = y$	Theorem 8.8, 8.9
Mixed Type	$x + K_1 N_{f_1} x + K_2 N_{f_2} x = y$	Superposition
Generalized	$u + K(u)N_f u = v$	Theorem 8.17
Urysohn	$x + Nx = y$	Direct method

Common Tools:

- Strong monotonicity of K ensures invertibility
- Monotonicity and boundedness of N_f enable fixed points
- Continuity and compactness assumptions ensure convergence
- Angle boundedness provides spectral localization

Python Example: Numerical Solution

```
#!/usr/bin/env python3
640: "page_641_hammerstein_integral",    # Integral Equations intro
```

Key Theorems Summary

- **Theorem 8.8 (Dolph-Minty):** Strongly monotone K + monotone $N_f \Rightarrow$ unique solution
- **Theorem 8.9:** Extension allowing unbounded N_f with growth conditions
- **Theorem 8.17:** Generalized Hammerstein with $K(u)$ depending on u
- **Corollary 8.3:** Urysohn equations with Carathéodory kernels
- **Theorem 8.23:** Leray-Schauder principles for approximate solvability

Common Hypothesis:

- 1 K is linear, continuous, monotone (or angle-bounded)
- 2 N_f is nonlinear, continuous, bounded, monotone
- 3 Suitable growth conditions on coefficients
- 4 Reflexivity of underlying Banach space

Applications and Extensions I

From Chapter 8

Physical Applications:

- Electrostatics and potential theory (kernels from Green's functions)
- Diffusion processes (parabolic PDEs via integral formulation)
- Mechanics and elasticity (stress-strain relationships)
- Quantum mechanics (scattering amplitudes)

Mathematical Extensions:

- Multivalued operators (inclusions vs. equations)
- Time-dependent kernels (parabolic equations)
- Fractional integral operators (singular kernels)
- Systems of integral equations

Computational Methods:

- Galerkin methods (discretization)

Applications and Extensions II

From Chapter 8

- Collocation schemes
- Boundary element methods (BEM)
- Iterative methods with convergence analysis

Open Problems:

- Nonsmooth integral operators
- Singular kernels with borderline regularity
- Coupled systems with memory effects
- Stochastic integral equations

References and Further Study

Main Reference:

- Pathak, H. K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer. Pages 641–664.

Related Topics:

- Chapter 2: Geometry of Banach spaces (monotone operators, convexity)
- Chapter 3: Nonexpansive mappings and fixed points
- Chapter 5: Fixed point theorems for various classes of mappings
- Chapter 6: Degree theory and topological methods

Classical References:

- Dolph-Minty theorem on monotone operators
- Hammerstein integral equations (classical theory)
- Leray-Schauder fixed point principle
- Backwinkel-Schillings method for generalized equations

Key Takeaways:

- 1 Integral equations naturally reduce to operator equations in Banach spaces
- 2 Monotonicity and angle-boundedness are crucial properties for solvability
- 3 Fixed point theory provides powerful tools for existence and uniqueness
- 4 Computational methods rely on finite-dimensional approximations and convergence analysis
- 5 Applications span across physics, mechanics, and modern analysis

Monotone Operator Theory: Bridge between Analysis and Computation